Practical: 11 Date:23-10-2026

Great Feature Engineering

Procedure:

1. Import the required Python libraries.
2. Load the Ames Housing dataset.
3. Display the dataset and check data types and missing values.
4. Plot graphs to study sale price distribution, correlations, and living area.
5. Separate the input features and target variable.
6. Handle missing values in the dataset.
7. Create new features such as total area, total bathrooms, and home quality.
8. Convert categorical variables into numerical form.
9. Split the data into training and validation sets.
10. Train the Random Forest Regression model.
11. Predict house sale prices using the trained model.
12. Evaluate the model using MAE, RMSE, and R-squared.
13. Plot actual versus predicted sale prices.
14. Identify and visualize the most important features.
15. Interpret the results and draw conclusions about the model's performance.

Code:

```
# Step 1: Import libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# Step 2: Load the Ames Housing dataset
train = pd.read_csv(r"E:\vscode\train.csv")
test = pd.read_csv(r"E:\vscode\test.csv")

print(train.head())
print(train.info())
print(train.isnull().sum())

# Graph 1: Distribution of SalePrice
fig, axes = plt.subplots(1, 2, figsize=(12, 5))
```

```
sns.histplot(train["SalePrice"], kde=True, ax=axes[0])
axes[0].set_title("Distribution of SalePrice")
axes[0].set_xlabel("Sale Price")

sns.histplot(np.log1p(train["SalePrice"]), kde=True, ax=axes[1])
axes[1].set_title("Log-Transformed SalePrice")
axes[1].set_xlabel("Log(SalePrice + 1)")

plt.tight_layout()
plt.show()

# Graph 2: Correlation heatmap
numeric_data = train.select_dtypes(include="number")

top_corr = (
    numeric_data.corr()["SalePrice"]
    .abs()
    .sort_values(ascending=False)
    .head(12)
    .index)

plt.figure(figsize=(10, 8))
sns.heatmap(
    train[top_corr].corr(),
    annot=True,
    cmap="coolwarm",
    fmt=".2f")
plt.title("Correlation Heatmap of Important Numerical Features")
plt.tight_layout()
plt.show()

# Graph 3: SalePrice vs Living Area
plt.figure(figsize=(8, 6))
sns.scatterplot(data=train, x="GrLivArea", y="SalePrice")
plt.title("SalePrice vs Above-Ground Living Area")
plt.xlabel("Living Area (square feet)")
plt.ylabel("Sale Price")
plt.tight_layout()
plt.show()

# Step 3: Separate target variable
y = train["SalePrice"]
X = train.drop(["SalePrice", "Id"], axis=1)
X_test = test.drop("Id", axis=1)

# Step 4: Combine train and test data for feature engineering
all_data = pd.concat([X, X_test], axis=0).reset_index(drop=True)
```

```
# Step 5: Convert categorical variables
for col in ["MSSubClass", "MoSold", "YrSold"]:
    all_data[col] = all_data[col].astype(str)

# Step 6: Handle missing values
categorical_cols = all_data.select_dtypes(
    include=["object", "str"]
).columns

numerical_cols = all_data.select_dtypes(
    exclude=["object", "str"]
).columns

for col in categorical_cols:
    all_data[col] = all_data[col].fillna("None")

for col in numerical_cols:
    all_data[col] = all_data[col].fillna(0)

# Step 7: Feature engineering
all_data["TotalSF"] = (
    all_data["TotalBsmtSF"]
    + all_data["1stFlrSF"]
    + all_data["2ndFlrSF"])

all_data["Total_Bathrooms"] = (
    all_data["FullBath"]
    + 0.5 * all_data["HalfBath"]
    + all_data["BsmtFullBath"]
    + 0.5 * all_data["BsmtHalfBath"])

all_data["Total_porch_sf"] = (
    all_data["OpenPorchSF"]
    + all_data["3SsnPorch"]
    + all_data["EnclosedPorch"]
    + all_data["ScreenPorch"]
    + all_data["WoodDeckSF"])

all_data["Total_Home_Quality"] = (
    all_data["OverallQual"] * all_data["GrLivArea"])

all_data["HasGarage"] = (all_data["GarageArea"] > 0).astype(int)
all_data["HasBsmt"] = (all_data["TotalBsmtSF"] > 0).astype(int)
all_data["HasFireplace"] = (all_data["Fireplaces"] > 0).astype(int)

# Step 8: Transform categorical variables into numerical variables
all_data = pd.get_dummies(
    all_data,
```

```
    columns=categorical_cols)
all_data = all_data.astype(float)

# Step 9: Split the data back into training and testing data
X = all_data.iloc[:len(train)].copy()
X_test = all_data.iloc[len(train):].copy()

X = X.fillna(0)
X_test = X_test.fillna(0)

# Step 10: Split training data into training and validation sets
X_train, X_valid, y_train, y_valid = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42)

# Step 11: Train Random Forest Regression model
model = RandomForestRegressor(
    n_estimators=500,
    max_depth=15,
    min_samples_split=5,
    min_samples_leaf=2,
    random_state=42,
    n_jobs=-1)

model.fit(X_train, y_train)

# Step 12: Make predictions
y_pred = model.predict(X_valid)

# Step 13: Evaluate model performance
mae = mean_absolute_error(y_valid, y_pred)
rmse = np.sqrt(mean_squared_error(y_valid, y_pred))
r2 = r2_score(y_valid, y_pred)
print("MAE :", mae)
print("RMSE:", rmse)
print("R²  :", r2)

# Graph 4: Actual vs Predicted Sale Prices
plt.figure(figsize=(8, 6))
plt.scatter(y_valid, y_pred, alpha=0.6)

limits = [
    min(y_valid.min(), y_pred.min()),
    max(y_valid.max(), y_pred.max())]

plt.plot(limits, limits, "r--")
plt.xlabel("Actual Sale Price")
plt.ylabel("Predicted Sale Price")
```

```
plt.title("Actual vs Predicted Sale Prices")
plt.tight_layout()
plt.show()

# Step 14: Feature importance
importance = pd.DataFrame({
    "Feature": X_train.columns,
    "Importance": model.feature_importances_})

importance = importance.sort_values(
    by="Importance",
    ascending=False)
print(importance.head(10))

# Graph 5: Top 15 Feature Importance
top_features = importance.head(15).sort_values(
    by="Importance",
    ascending=True)

plt.figure(figsize=(10, 7))
plt.barh(
    top_features["Feature"],
    top_features["Importance"])

plt.xlabel("Importance")
plt.ylabel("Feature")
plt.title("Top 15 Important Features")
plt.tight_layout()
plt.show()

# Step 1: Separate the target variable
y = train["SalePrice"]
X = train.drop(["SalePrice", "Id"], axis=1)

# Step 2: Convert categorical columns into numerical columns
X = pd.get_dummies(X, dummy_na=True)
X = X.astype(float)

# Step 3: Fill missing values
X = X.fillna(X.median())

# Step 4: Split the data
X_train, X_valid, y_train, y_valid = train_test_split(
    X, y, test_size=0.20, random_state=42)

# Step 5: Train the Random Forest model
model = RandomForestRegressor(
    n_estimators=200,
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   random_state=42,
    n_jobs=-1)

model.fit(X_train, y_train)

# Step 6: Predict house prices
y_pred = model.predict(X_valid)

# Step 7: Evaluate the model
mae = mean_absolute_error(y_valid, y_pred)
rmse = np.sqrt(mean_squared_error(y_valid, y_pred))
r2 = r2_score(y_valid, y_pred)

print("\n--- MODEL EVALUATION ---")
print("MAE:", mae)
print("RMSE:", rmse)
print("R-squared:", r2)
```

Output:

```
Id  MSSubClass MSZoning  ...  SaleType SaleCondition SalePrice

0   1          60        RL  ...       WD        Normal    208500

1   2          20        RL  ...       WD        Normal    181500

2   3          60        RL  ...       WD        Normal    223500

3   4          70        RL  ...       WD       Abnorml    140000

4   5          60        RL  ...       WD        Normal    250000


[5 rows x 81 columns]

<class 'pandas.DataFrame'>

RangeIndex: 1460 entries, 0 to 1459

Data columns (total 81 columns):

 #   Column         Non-Null Count  Dtype

---  ------         --------------  -----

 0   Id             1460 non-null   int64

 1   MSSubClass     1460 non-null   int64
```

2 MSZoning 1460 non-null str

3 LotFrontage 1201 non-null float64

4 LotArea 1460 non-null int64

5 Street 1460 non-null str

6 Alley 91 non-null str

7 LotShape 1460 non-null str

8 LandContour 1460 non-null str

9 Utilities 1460 non-null str

10 LotConfig 1460 non-null str

11 LandSlope 1460 non-null str

12 Neighborhood 1460 non-null str

13 Condition1 1460 non-null str

14 Condition2 1460 non-null str

15 BldgType 1460 non-null str

16 HouseStyle 1460 non-null str

17 OverallQual 1460 non-null int64

18 OverallCond 1460 non-null int64

19 YearBuilt 1460 non-null int64

20 YearRemodAdd 1460 non-null int64

21 RoofStyle 1460 non-null str

22 RoofMatl 1460 non-null str

23 Exterior1st 1460 non-null str

24 Exterior2nd 1460 non-null str

25 MasVnrType 588 non-null str

26 MasVnrArea 1452 non-null float64

27 ExterQual 1460 non-null str

28 ExterCond 1460 non-null str

29 Foundation 1460 non-null str

30 BsmtQual 1423 non-null str

31 BsmtCond 1423 non-null str

32 BsmtExposure 1422 non-null str

33 BsmtFinType1 1423 non-null str

34 BsmtFinSF1 1460 non-null int64

35 BsmtFinType2 1422 non-null str

36 BsmtFinSF2 1460 non-null int64

37 BsmtUnfSF 1460 non-null int64

38 TotalBsmtSF 1460 non-null int64

39 Heating 1460 non-null str

40 HeatingQC 1460 non-null str

41 CentralAir 1460 non-null str

42 Electrical 1459 non-null str

43 1stFlrSF 1460 non-null int64

44 2ndFlrSF 1460 non-null int64

45 LowQualFinSF 1460 non-null int64

46 GrLivArea 1460 non-null int64

47 BsmtFullBath 1460 non-null int64

48 BsmtHalfBath 1460 non-null int64

49 FullBath 1460 non-null int64

50 HalfBath 1460 non-null int64

51 BedroomAbvGr 1460 non-null int64

52 KitchenAbvGr 1460 non-null int64

53 KitchenQual 1460 non-null str

54 TotRmsAbvGrd 1460 non-null int64

55 Functional 1460 non-null str

56 Fireplaces 1460 non-null int64

57 FireplaceQu 770 non-null str

58 GarageType 1379 non-null str

59 GarageYrBlt 1379 non-null float64

60 GarageFinish 1379 non-null str

61 GarageCars 1460 non-null int64

62 GarageArea 1460 non-null int64

63 GarageQual 1379 non-null str

64 GarageCond 1379 non-null str

65 PavedDrive 1460 non-null str

66 WoodDeckSF 1460 non-null int64

67 OpenPorchSF 1460 non-null int64

68 EnclosedPorch 1460 non-null int64

69 3SsnPorch 1460 non-null int64

70 ScreenPorch 1460 non-null int64

71 PoolArea 1460 non-null int64

72 PoolQC 7 non-null str

73 Fence 281 non-null str

74 MiscFeature 54 non-null str

75 MiscVal 1460 non-null int64

76 MoSold 1460 non-null int64

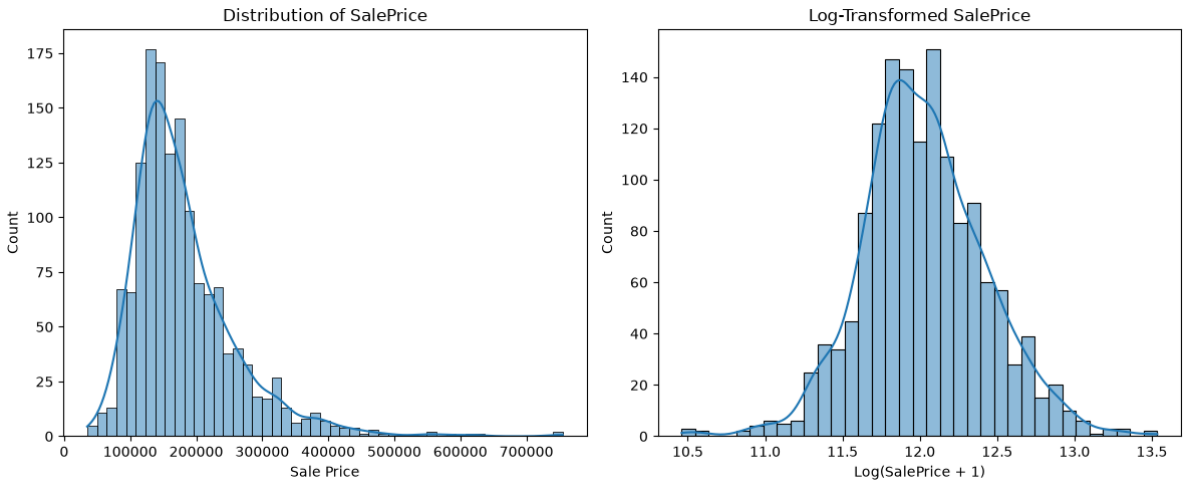
77 YrSold 1460 non-null int64

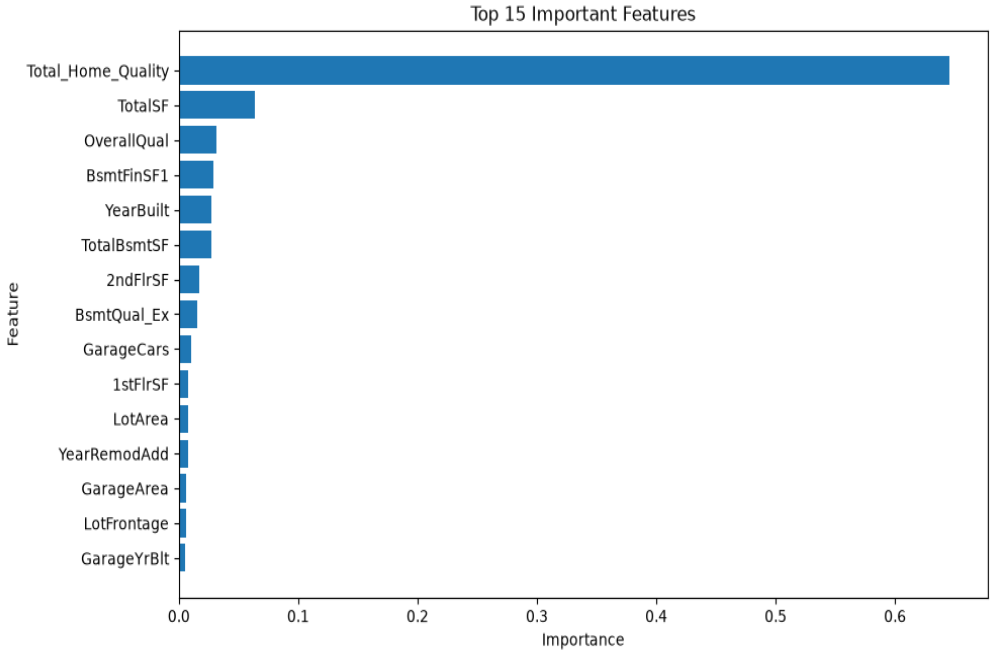
78 SaleType 1460 non-null str

79 SaleCondition 1460 non-null str

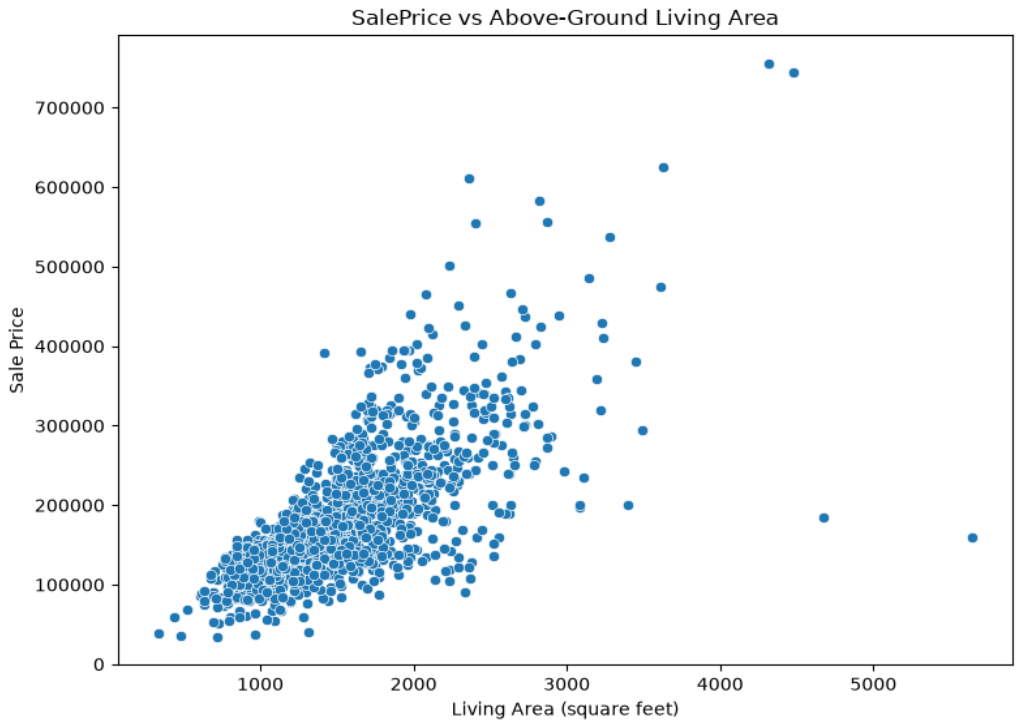
```
 80  SalePrice      1460 non-null   int64
dtypes: float64(3), int64(35), str(43)
memory usage: 924.0 KB
None
Id               0
MSSubClass       0
MSZoning         0
LotFrontage    259
LotArea          0
                ...
MoSold           0
YrSold           0
SaleType         0
SaleCondition    0
SalePrice        0
Length: 81, dtype: int64

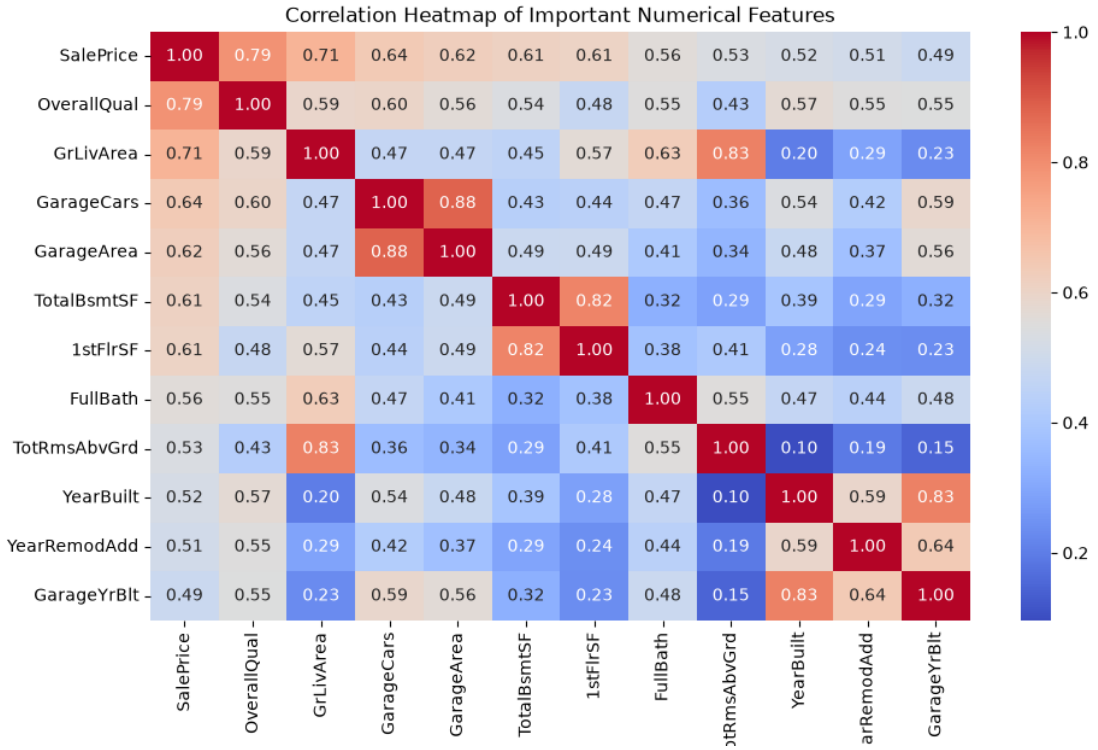
--- MODEL EVALUATION ---
MAE: 17573.68772260274
RMSE: 28674.073724070742
R-squared: 0.8928073615483214
```


Distribution of SalePrice
Count
Sale Price
0
25
50
75
100
125
150
175
0
100000
200000
300000
400000
500000
600000
700000
Log-Transformed SalePrice
Count
Log(SalePrice + 1)
0
20
40
60
80
100
120
140
10.5
11.0
11.5
12.0
12.5
13.0
13.5


Top 15 Important Features
Total_Home_Quality
TotalSF
OverallQual
BsmtFinSF1
YearBuilt
TotalBsmtSF
2ndFlrSF
BsmtQual_Ex
GarageCars
1stFlrSF
LotArea
YearRemodAdd
GarageArea
LotFrontage
GarageYrBlt
Feature
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Importance


Actual vs Predicted Sale Prices
Predicted Sale Price
0
100000
200000
300000
400000
500000
600000
700000
Actual Sale Price


SalePrice vs Above-Ground Living Area
Sale Price
Living Area (square feet)
0
100000
200000
300000
400000
500000
600000
700000
1000
2000
3000
4000
5000


Correlation Heatmap of Important Numerical Features
SalePrice 1.00 0.79 0.71 0.64 0.62 0.61 0.61 0.56 0.53 0.52 0.51 0.49
OverallQual 0.79 1.00 0.59 0.60 0.56 0.54 0.48 0.55 0.43 0.57 0.55 0.55
GrLivArea 0.71 0.59 1.00 0.47 0.47 0.45 0.57 0.63 0.83 0.20 0.29 0.23
GarageCars 0.64 0.60 0.47 1.00 0.88 0.43 0.44 0.47 0.36 0.54 0.42 0.59
GarageArea 0.62 0.56 0.47 0.88 1.00 0.49 0.49 0.41 0.34 0.48 0.37 0.56
TotalBsmtSF 0.61 0.54 0.45 0.43 0.49 1.00 0.82 0.32 0.29 0.39 0.29 0.32
1stFlrSF 0.61 0.48 0.57 0.44 0.49 0.82 1.00 0.38 0.41 0.28 0.24 0.23
FullBath 0.56 0.55 0.63 0.47 0.41 0.32 0.38 1.00 0.55 0.47 0.44 0.48
TotRmsAbvGrd 0.53 0.43 0.83 0.36 0.34 0.29 0.41 0.55 1.00 0.10 0.19 0.15
YearBuilt 0.52 0.57 0.20 0.54 0.48 0.39 0.28 0.47 0.10 1.00 0.59 0.83
YearRemodAdd 0.51 0.55 0.29 0.42 0.37 0.29 0.24 0.44 0.19 0.59 1.00 0.64
GarageYrBlt 0.49 0.55 0.23 0.59 0.56 0.32 0.23 0.48 0.15 0.83 0.64 1.00
SalePrice
OverallQual
GrLivArea
GarageCars
GarageArea
TotalBsmtSF
1stFlrSF
FullBath
YearBuilt
GarageYrBlt
1.0
0.8
0.6
0.4
0.2